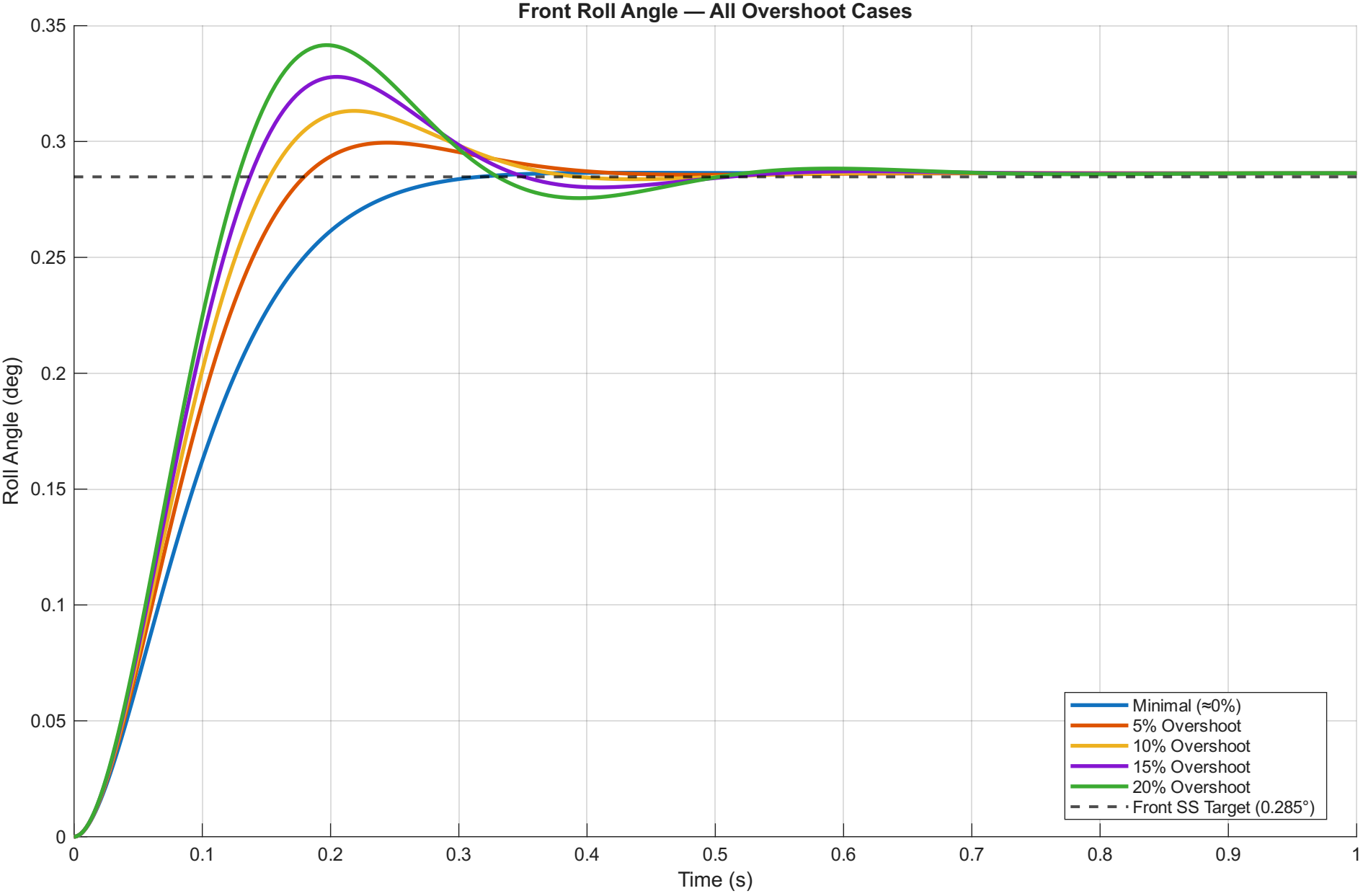
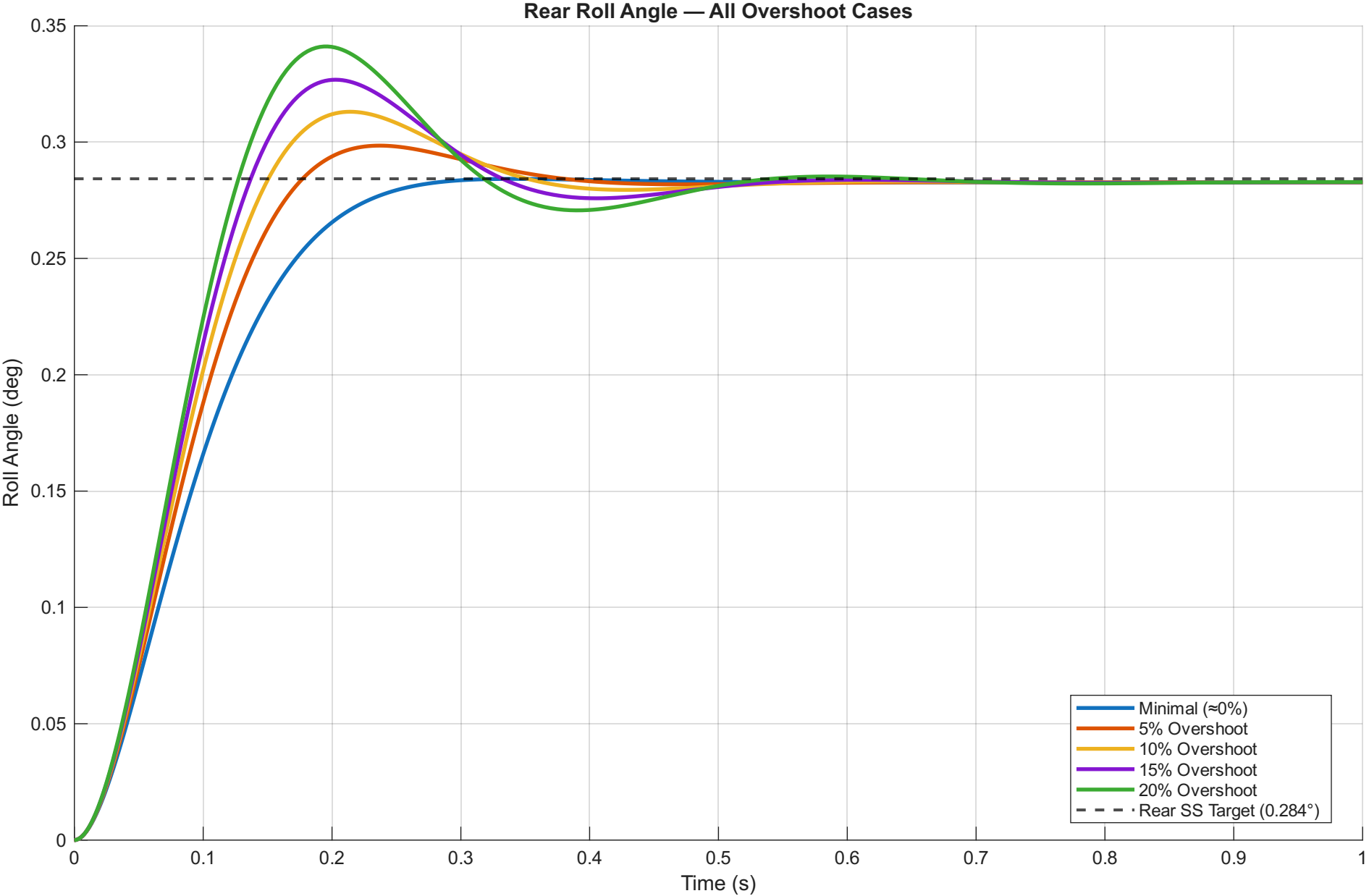
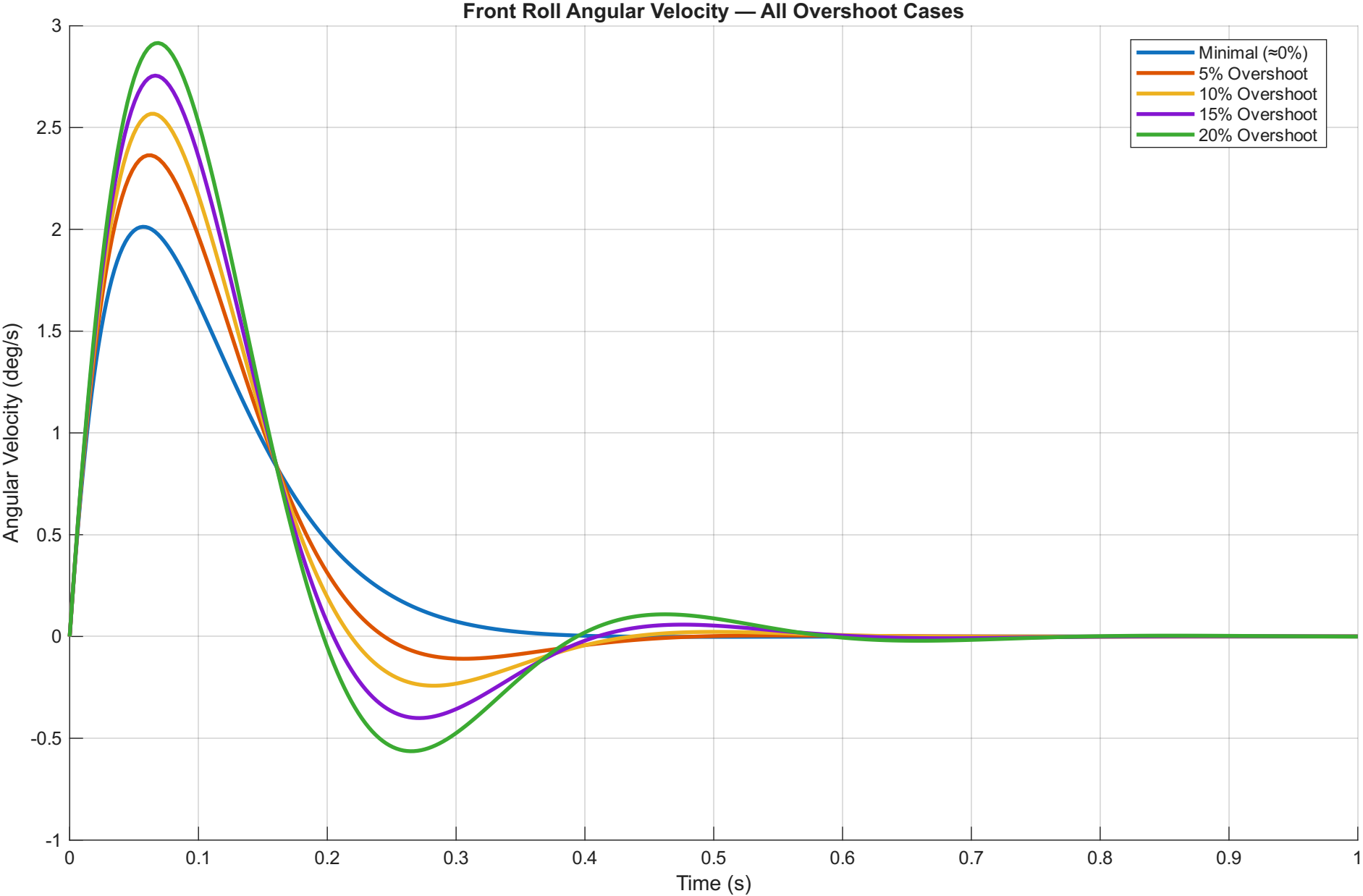
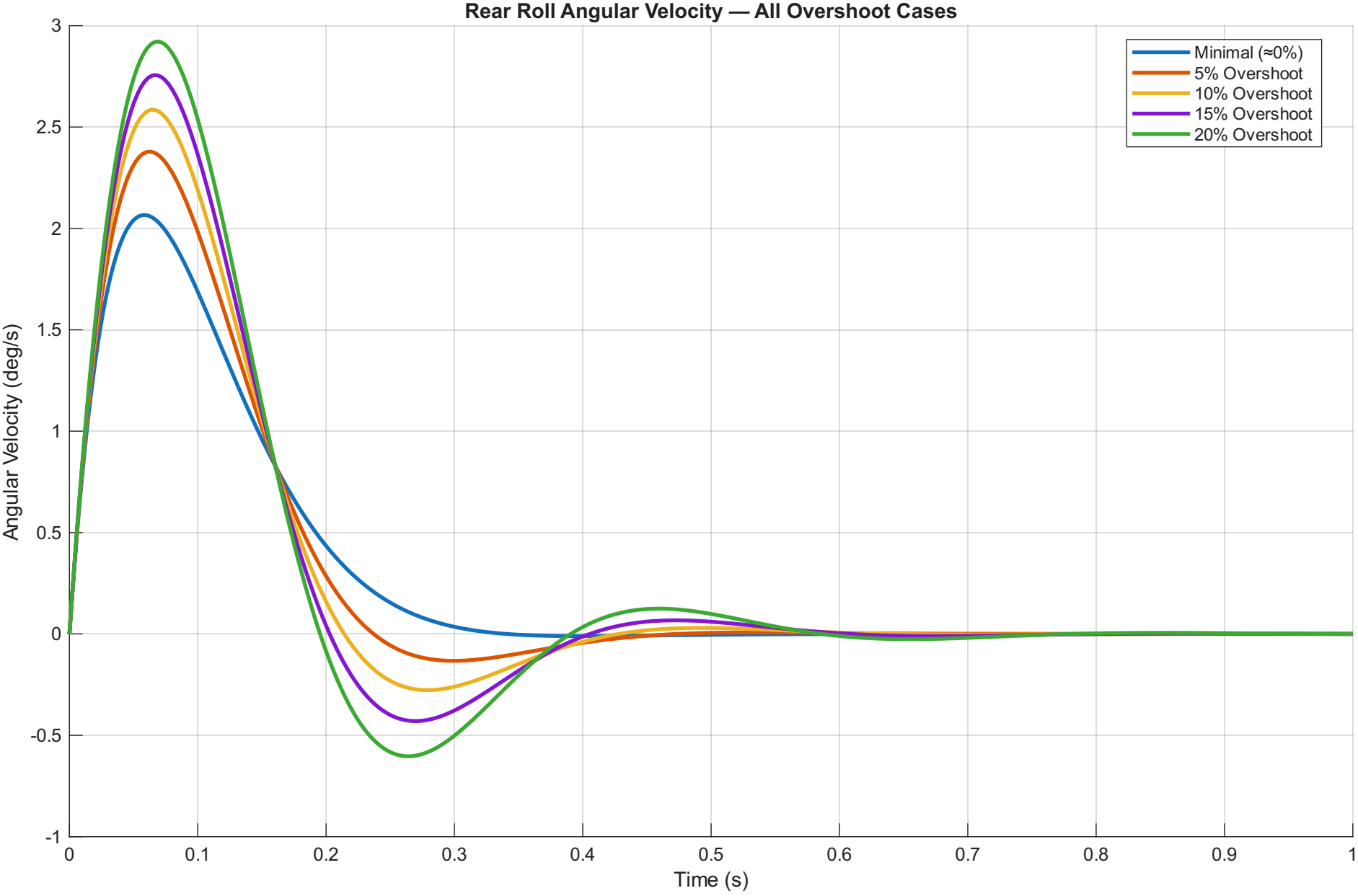
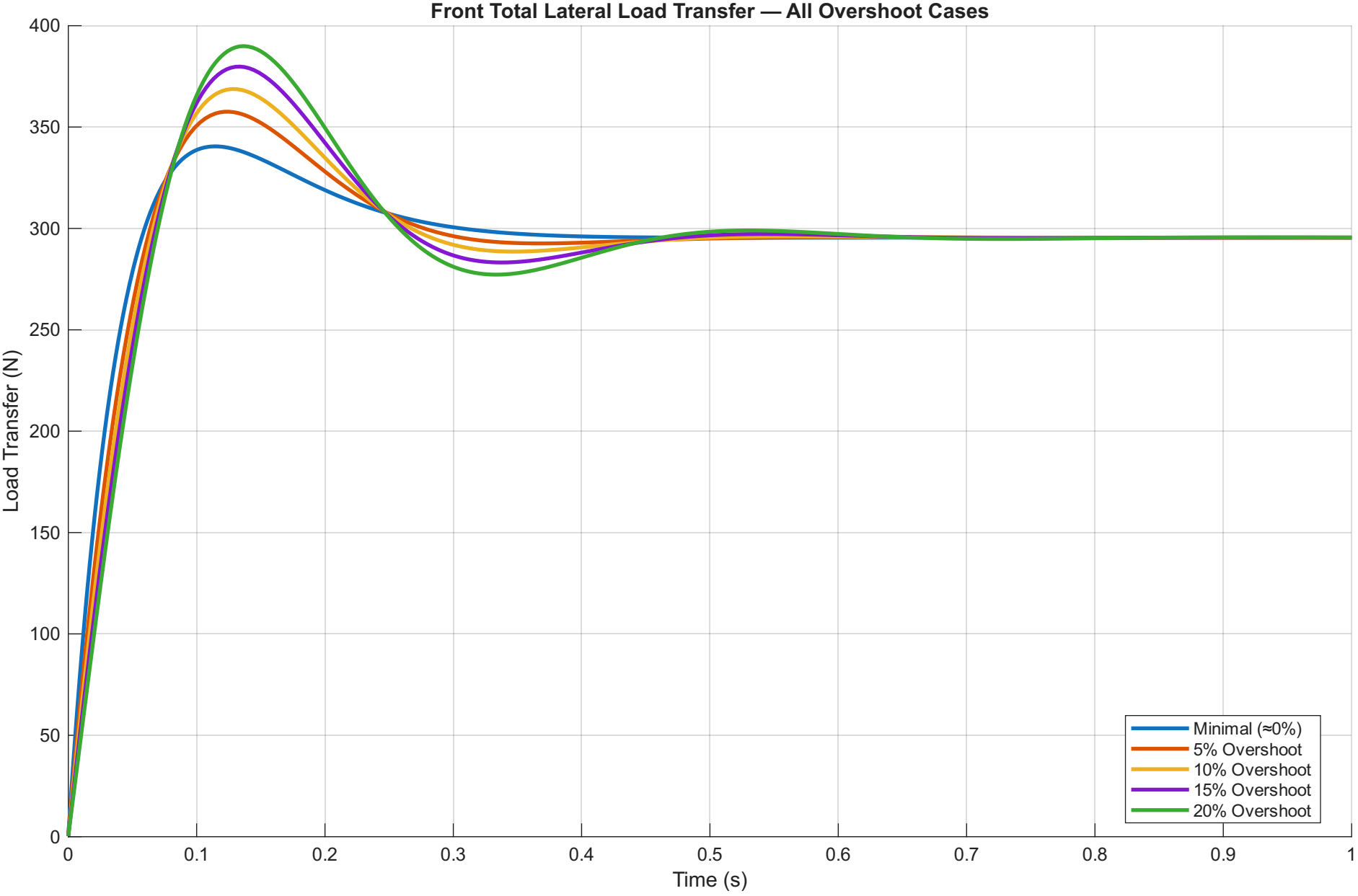
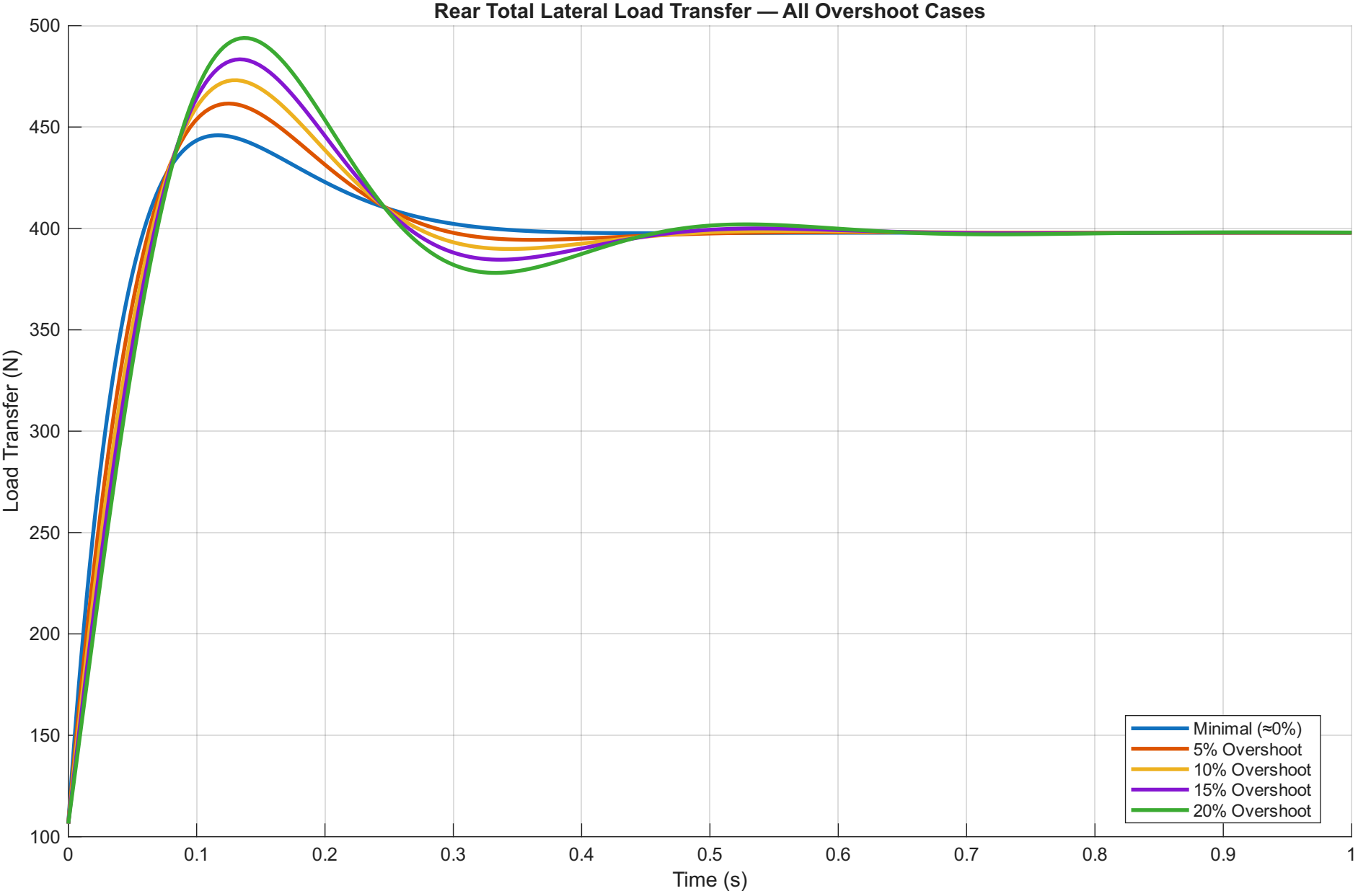

Front Roll Angle — All Overshoot Cases
Roll Angle (deg)
Time (s)
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Minimal (≈0%)
5% Overshoot
10% Overshoot
15% Overshoot
20% Overshoot
Front SS Target (0.285°)

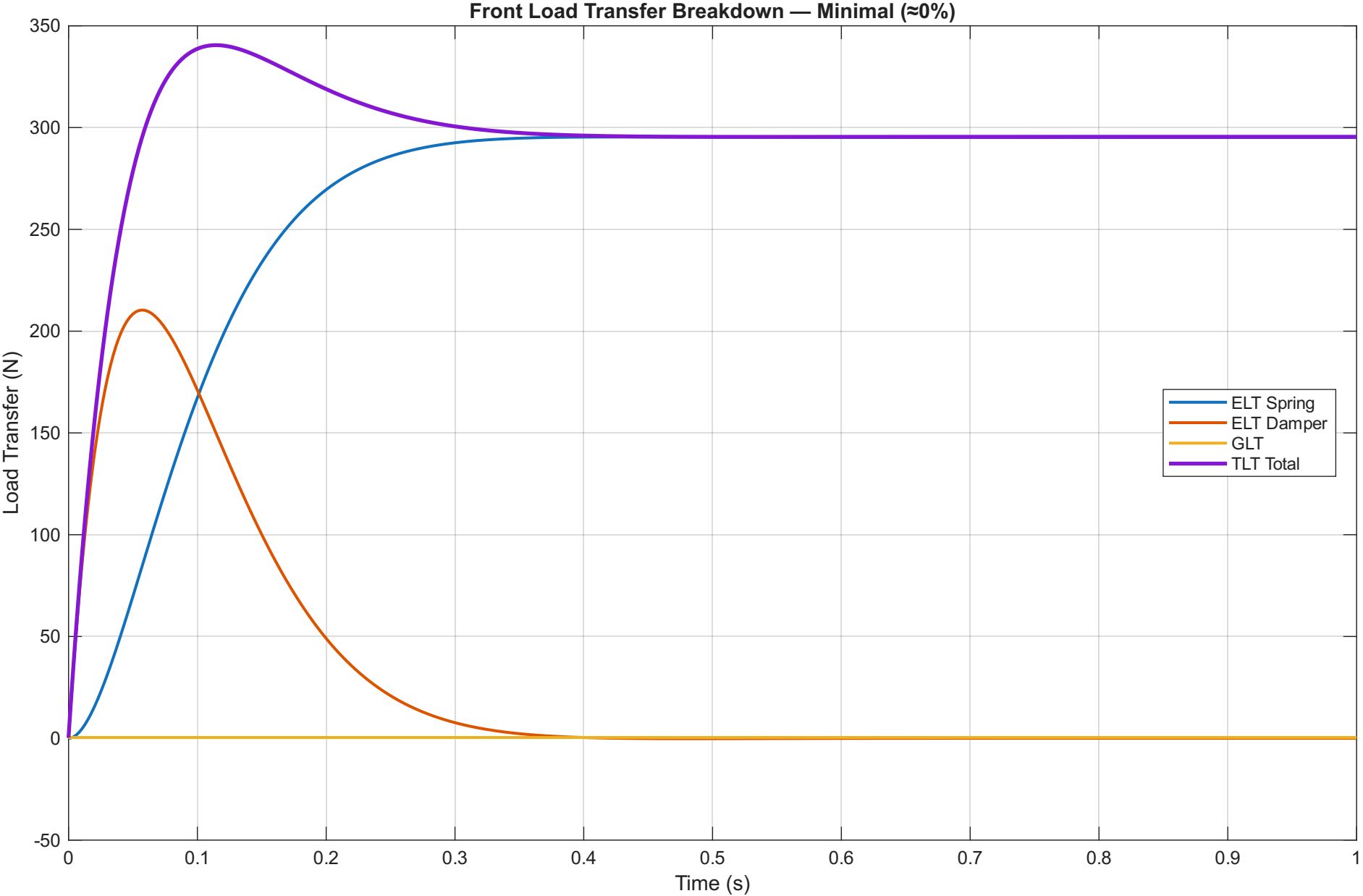

Rear Roll Angle — All Overshoot Cases
Roll Angle (deg)
Time (s)
0.35
0.3
0.25
0.2
0.15
0.1
0.05
0
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Minimal (≈0%)
5% Overshoot
10% Overshoot
15% Overshoot
20% Overshoot
Rear SS Target (0.284°)

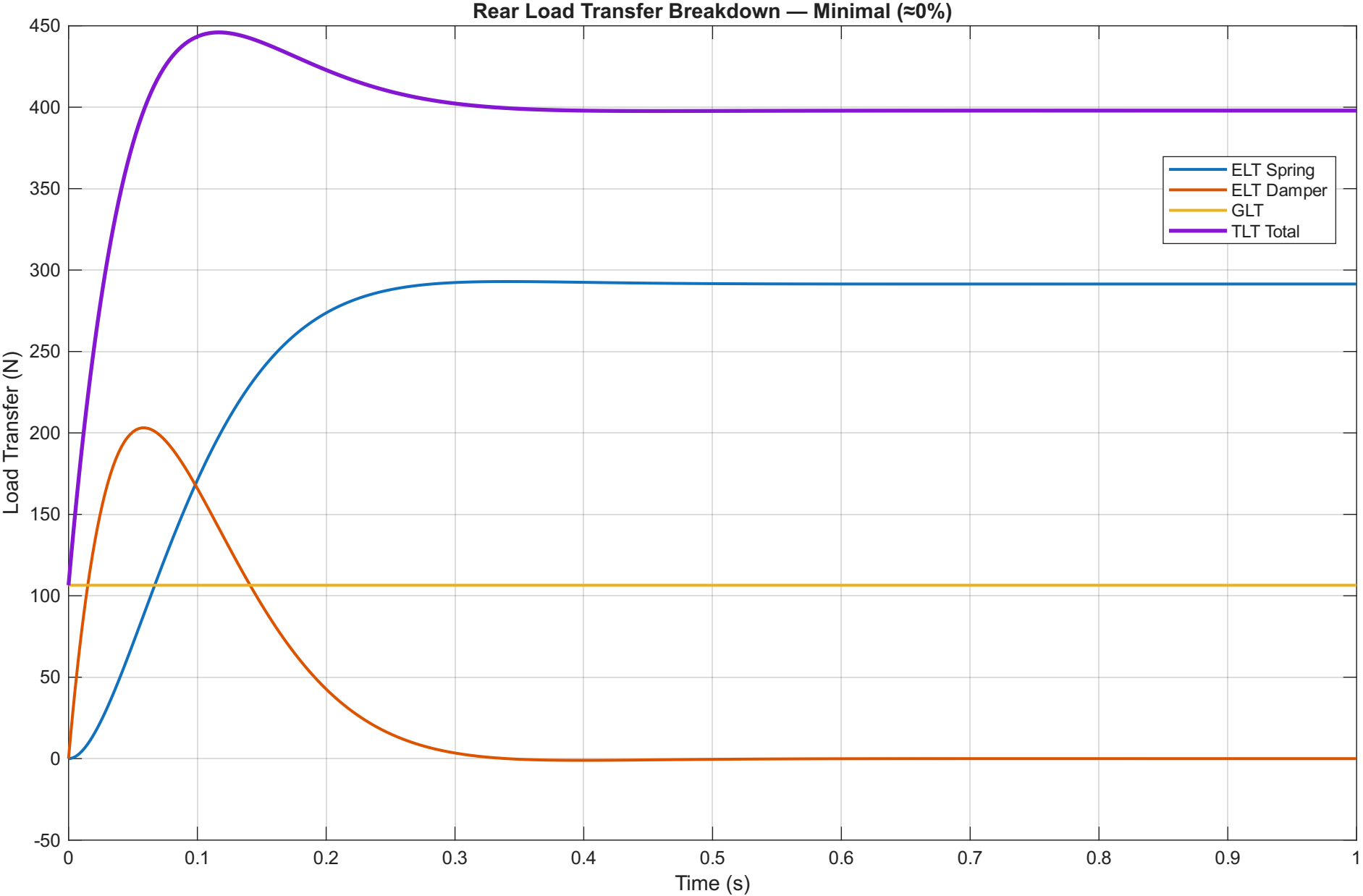

Front Roll Angular Velocity — All Overshoot Cases
Minimal (≈0%)
5% Overshoot
10% Overshoot
15% Overshoot
20% Overshoot
Angular Velocity (deg/s)
Time (s)
3
2.5
2
1.5
1
0.5
0
-0.5
-1
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1

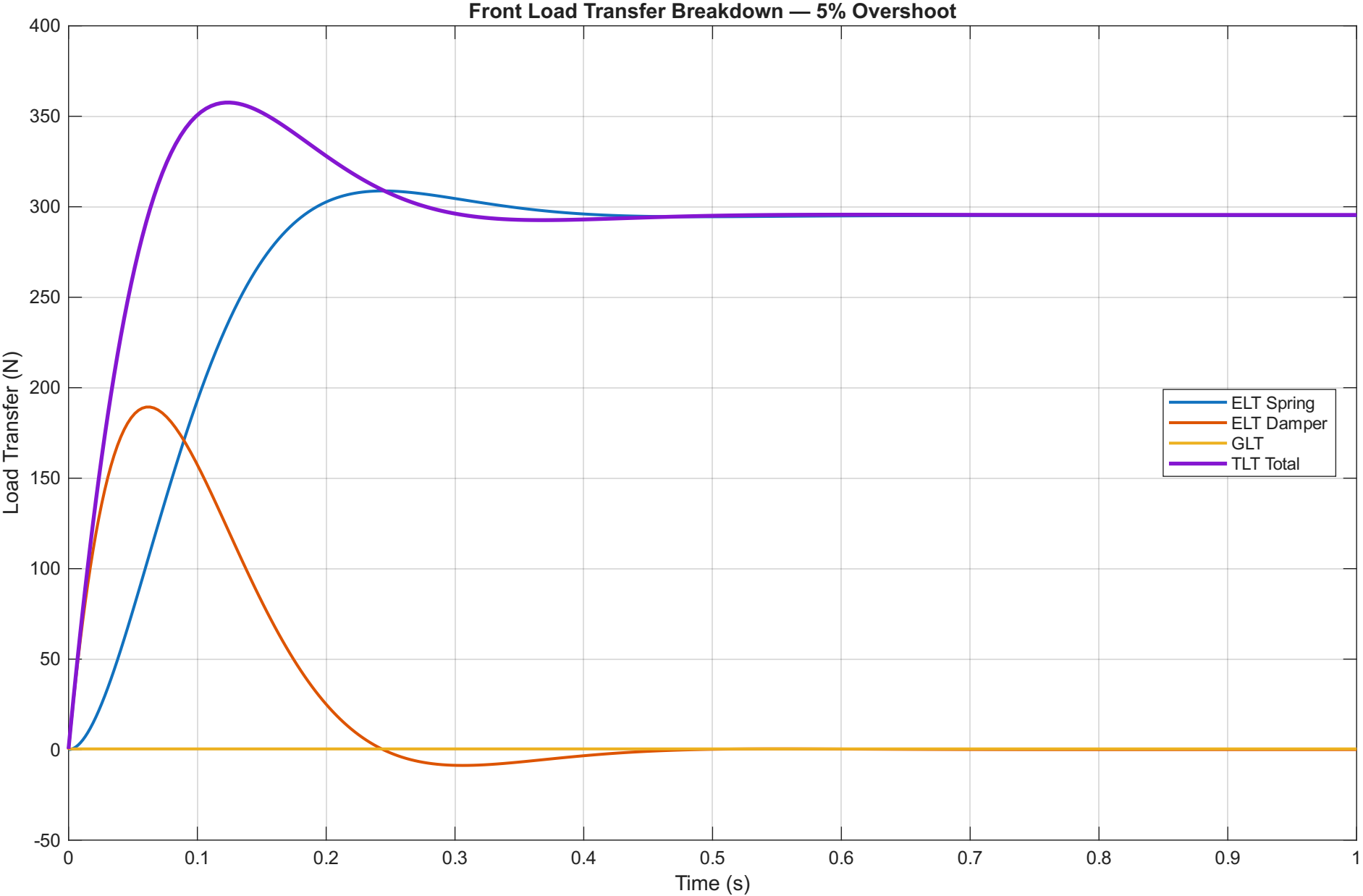

Rear Roll Angular Velocity — All Overshoot Cases
Minimal (≈0%)
5% Overshoot
10% Overshoot
15% Overshoot
20% Overshoot
Angular Velocity (deg/s)
Time (s)
3
2.5
2
1.5
1
0.5
0
-0.5
-1
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1

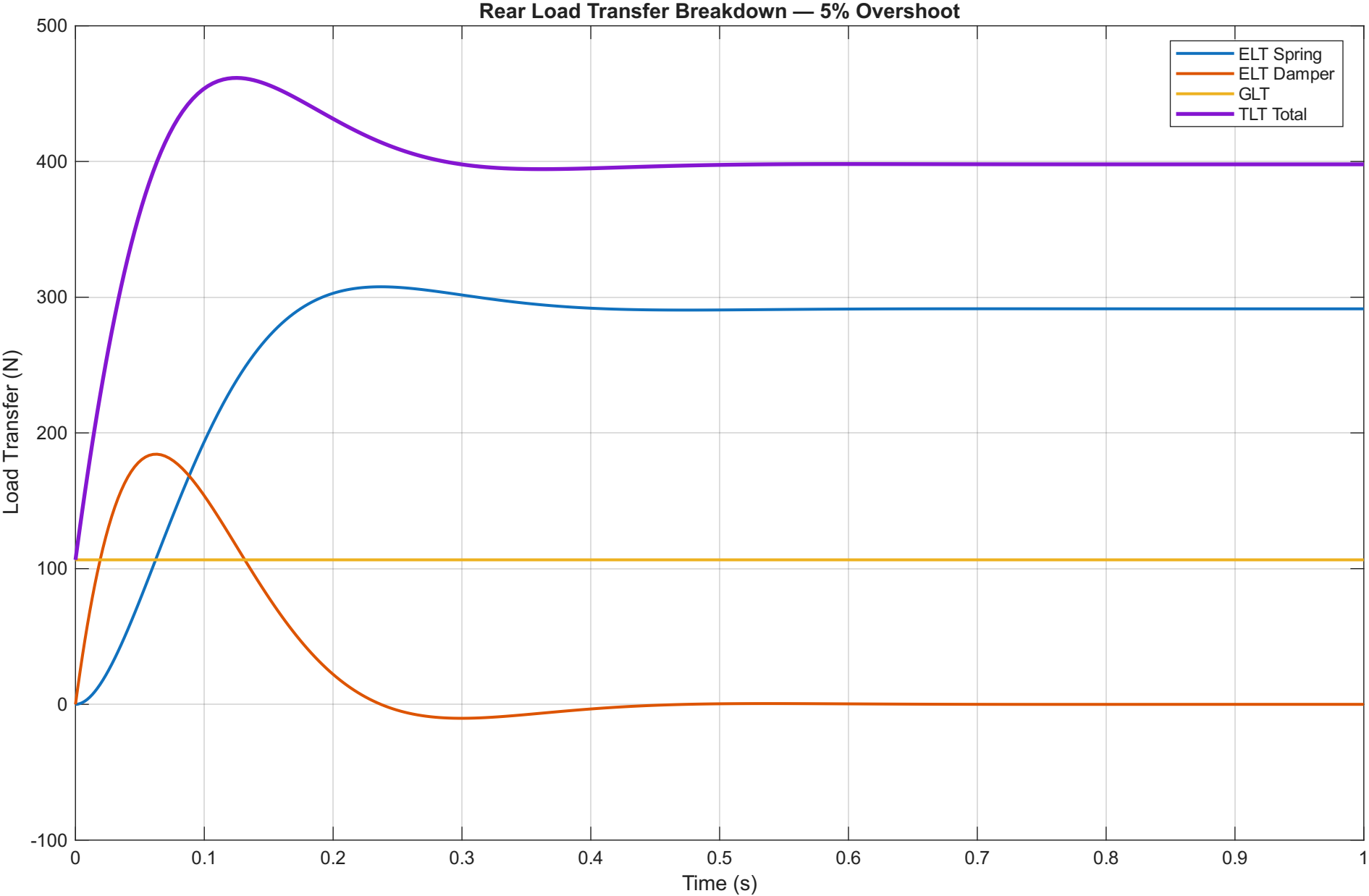

Front Total Lateral Load Transfer — All Overshoot Cases
Load Transfer (N)
Time (s)
0
50
100
150
200
250
300
350
400
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Minimal (≈0%)
5% Overshoot
10% Overshoot
15% Overshoot
20% Overshoot

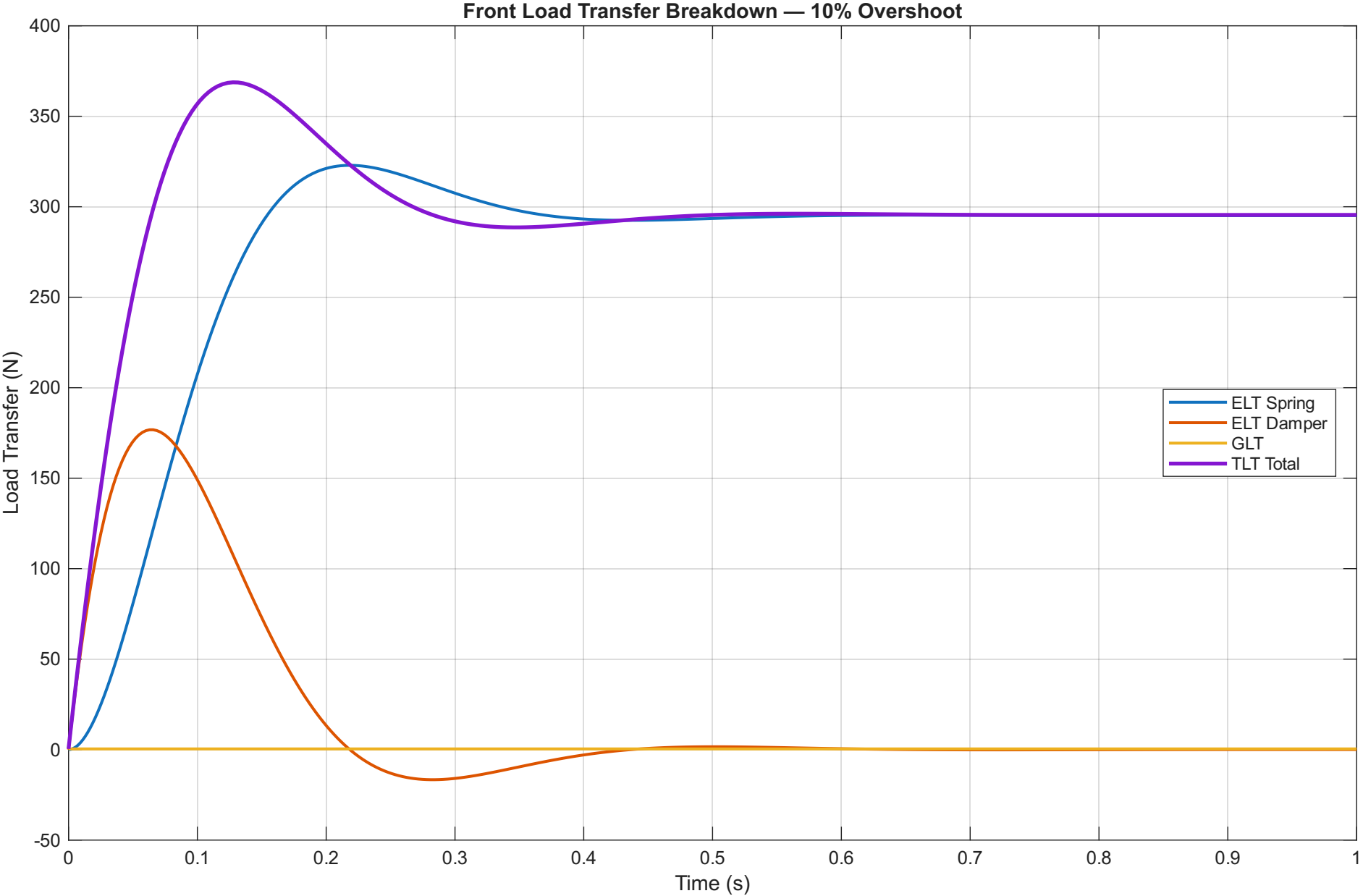

Rear Total Lateral Load Transfer — All Overshoot Cases
Load Transfer (N)
Time (s)
500
450
400
350
300
250
200
150
100
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Minimal (≈0%)
5% Overshoot
10% Overshoot
15% Overshoot
20% Overshoot

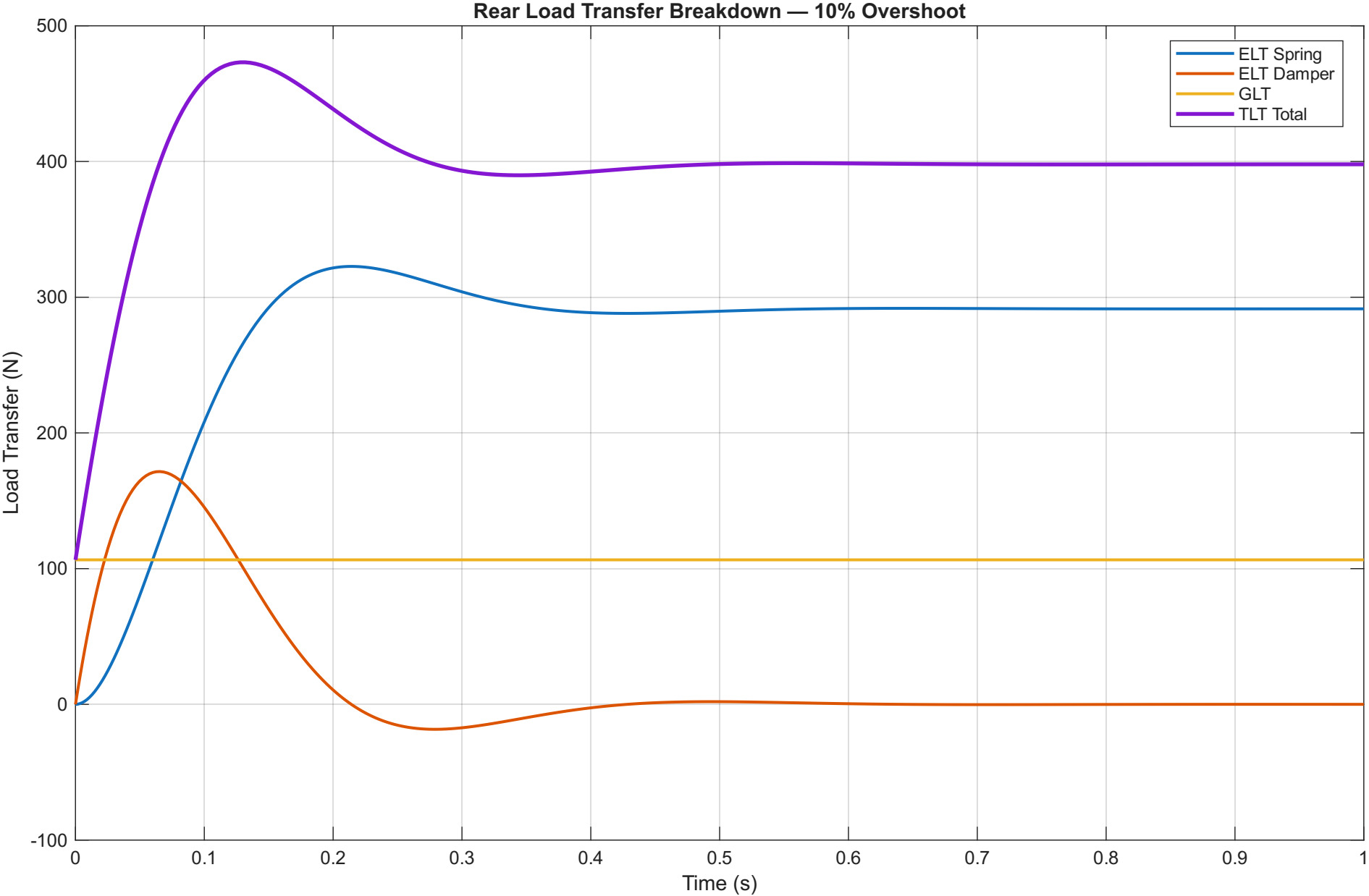

Front Load Transfer Breakdown — Minimal (≈0%)
Load Transfer (N)
Time (s)
350
300
250
200
150
100
50
0
-50
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
ELT Spring
ELT Damper
GLT
TLT Total

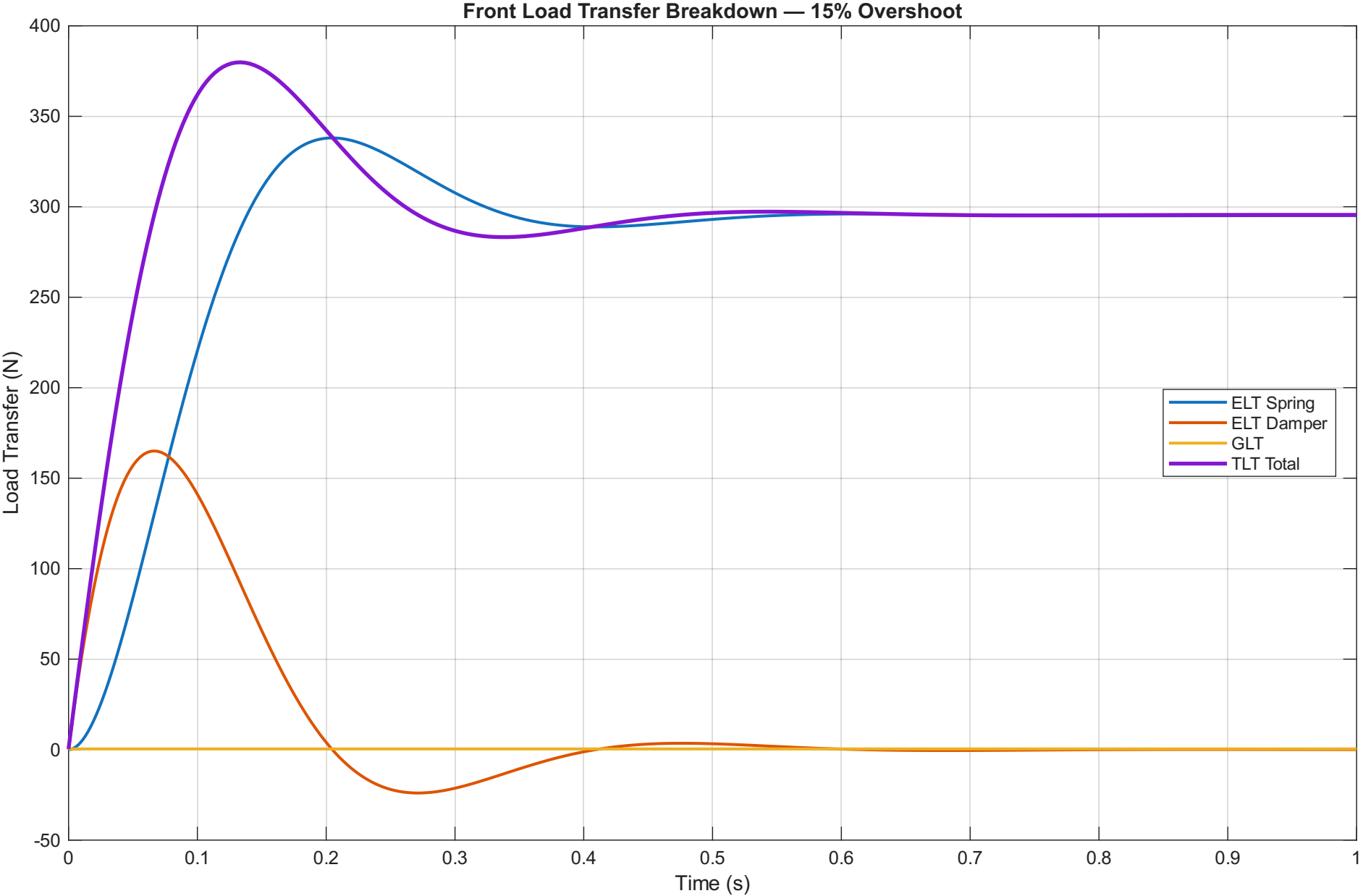

Rear Load Transfer Breakdown — Minimal (≈0%)
Load Transfer (N)
Time (s)
ELT Spring
ELT Damper
GLT
TLT Total
450
400
350
300
250
200
150
100
50
0
-50
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1

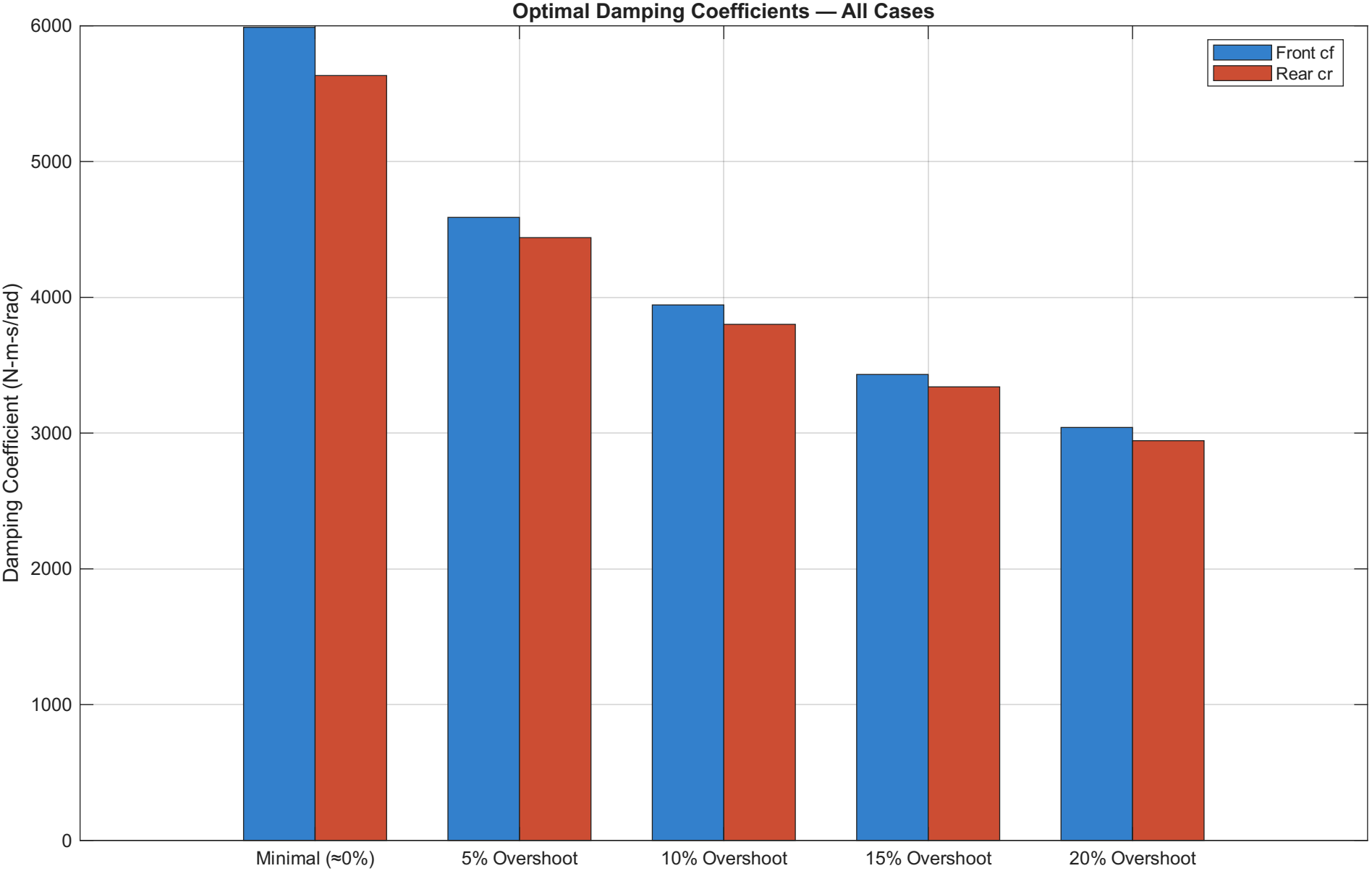

Front Load Transfer Breakdown — 5% Overshoot
Load Transfer (N)
Time (s)
400
350
300
250
200
150
100
50
0
-50
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
ELT Spring
ELT Damper
GLT
TLT Total

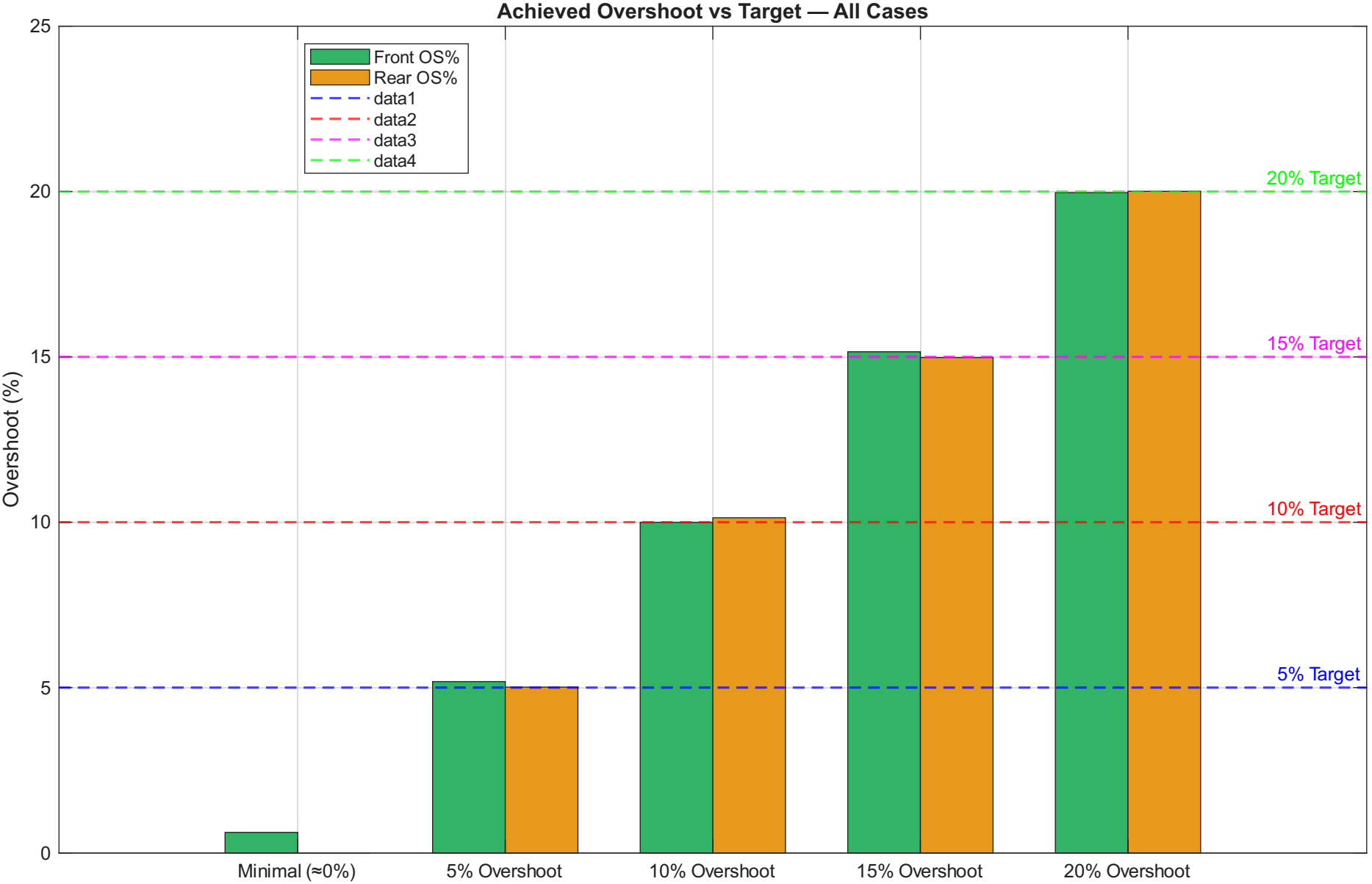
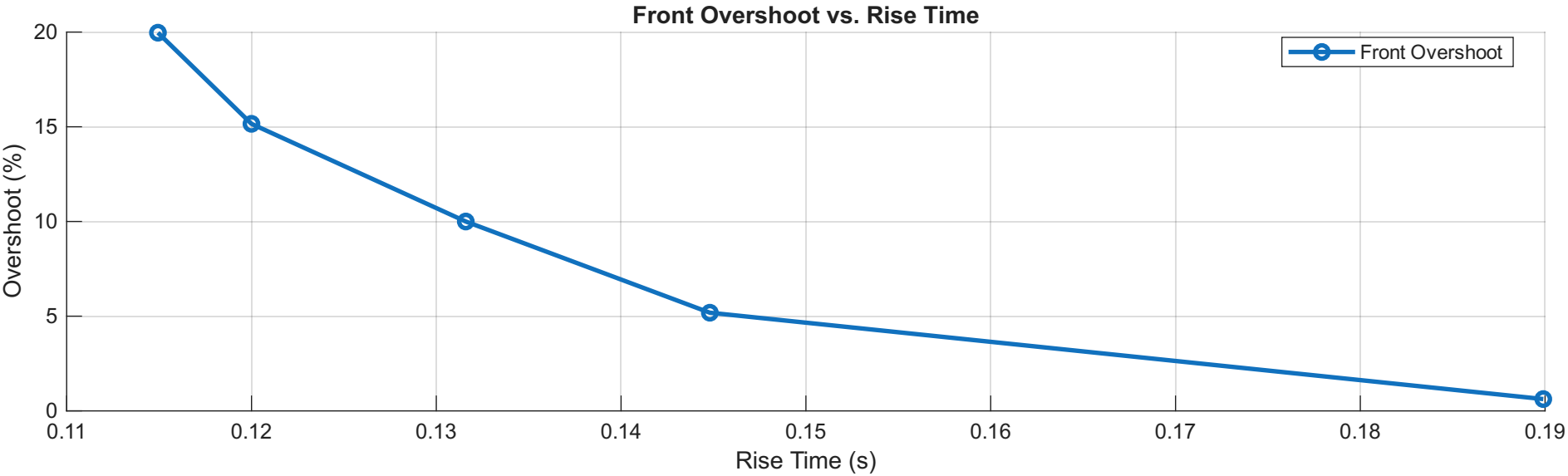
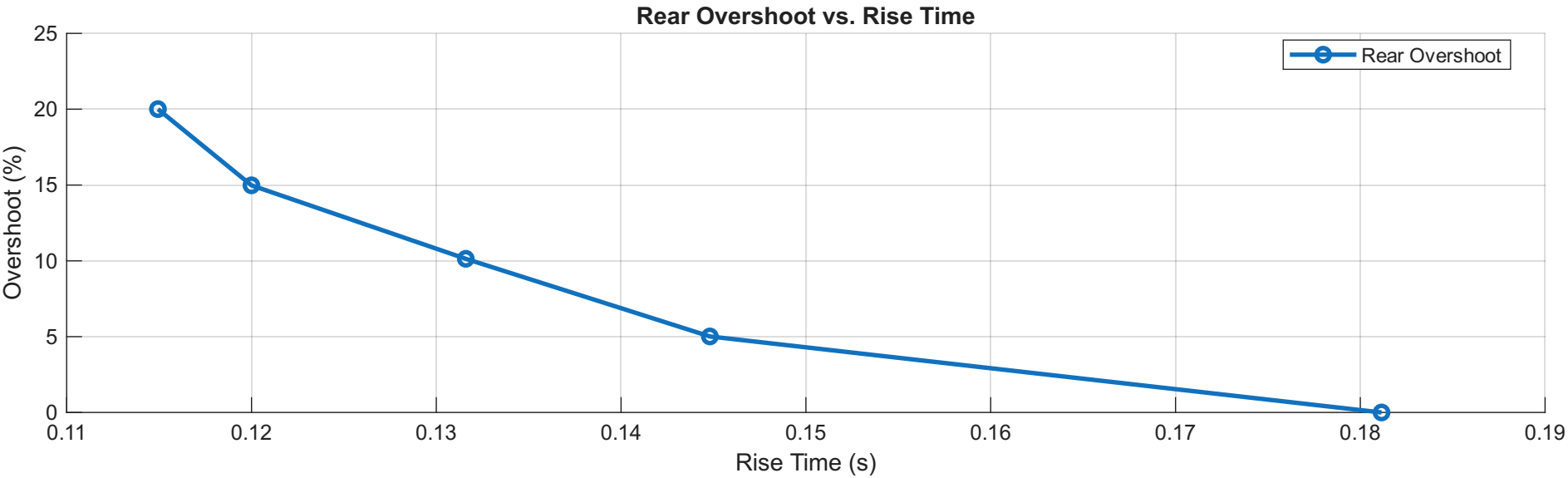

Rear Load Transfer Breakdown — 5% Overshoot
ELT Spring
ELT Damper
GLT
TLT Total
Load Transfer (N)
Time (s)
500
400
300
200
100
0
-100
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1


Front Load Transfer Breakdown — 10% Overshoot
Load Transfer (N)
Time (s)
400
350
300
250
200
150
100
50
0
-50
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
ELT Spring
ELT Damper
GLT
TLT Total


Rear Load Transfer Breakdown — 10% Overshoot
ELT Spring
ELT Damper
GLT
TLT Total
Load Transfer (N)
Time (s)
500
400
300
200
100
0
-100
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1


Front Load Transfer Breakdown — 15% Overshoot
Load Transfer (N)
Time (s)
400
350
300
250
200
150
100
50
0
-50
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
ELT Spring
ELT Damper
GLT
TLT Total


Optimal Damping Coefficients — All Cases
Damping Coefficient (N-m-s/rad)
6000
5000
4000
3000
2000
1000
0
Front cf
Rear cr
Minimal (≈0%)
5% Overshoot
10% Overshoot
15% Overshoot
20% Overshoot


Achieved Overshoot vs Target — All Cases
Front OS%
Rear OS%
data1
data2
data3
data4
20% Target
15% Target
10% Target
5% Target
Overshoot (%)
25
20
15
10
5
0
Minimal (≈0%)
5% Overshoot
10% Overshoot
15% Overshoot
20% Overshoot

Sensitivity Study: Overshoot Percentage vs. Rise Time